CARBON ZAPP

INNOVATION DRIVEN

Engineering Forward Mobility Solutions!

CTBR „X

The Compact Diesel CRDi Test Bench

Bench-top Specialist Service Station. Compact Footprint. Full CRDi Capability.



Authorized Solution for
Schaeffler / Continental

Continental

CloudX AZO

CRDi INJECTORS

CTBR X

Compact CRDi Power. Bench-Top Footprint.

Full common-rail testing and coding, every brand, up to 2,500 bar, every EU6 technology, in a compact bench-top that runs on single-phase power. No three-phase install, no dedicated floor space.

Brand	Families & Applications
Bosch	CRi1, CRi2, CRi3, CRiN1, CRiN2, CRiN3 – passenger car, light-duty, medium-duty
Denso	G2, G3, G4 platforms – passenger car and medium-duty
Delphi	DFI 1.x, 2.x, 3.x (C2i / C3i) – passenger car and light-duty
Schaeffler / Continental	Piezo CRi injectors – full coverage with CGL Blind-Lift
Heavy-Duty	XPI, F2X / F3X, CRIN 4.2



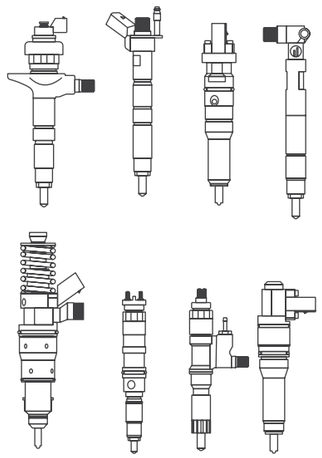
Pressure up to 2,500 bar

Database 4,500+ part numbers

continuously online-updated

EU6 technologies

VCC, NCC, iART, PCRS5



CRDi · COMMON-RAIL DIESEL INJECTORS

Universal CRDi injector testing and coding for every major diesel injector platform in production, passenger car, light-duty and medium-duty (Heavy-Duty via the Exclusive Version upgrade).



FEATURES

- Tested to OEM Standards
- Testing and coding time: 8–14 min
- **Heavy-Duty upgrade:** CRIN 4.2 (optional)
- Full injector coding: Bosch IQA/IMA, Denso QR, Delphi C2i/C3i, Schaeffler / Continental IIC, CAT (TRM)
- CGL: authorized Schaeffler / Continental “Blind-Lift” Measurement
- Measurement of piezo-stack
- Live response-time measurement at every test plan
- Single-phase 50/60 Hz VFD motor-driven: no air supply required

TESTS PERFORMED

- Electrical (coil / piezo)
- Insulation
- Back-leak (static / dynamic)
- Volumetric flow (all conditions)
- Nozzle opening pressure (aNOP)
- Opening & closing response time (RSP – eRSP)
- EU6 technology testing
- DFI mapping (full spectrum)
- NLT / LKT nozzle leak test
- Spray pattern & atomization
- RDS rail-pressure sensors (all brands)
- HP rails (without DRV)

AZO

The intelligence inside every Carbon Zapp machine.

Every Carbon Zapp X-Series bench is driven by AZO, the software platform at the heart of the machine's powerful touchscreen HMI. AZO turns complex injection diagnostics into a guided, fully automatic process: select the component, mount it, and let the machine run the complete OE-level test plan, from actuation and leak checks to dynamic precision flow measurement and coding. The result is expert-level diagnosis, delivered consistently, regardless of who is operating the bench.



AZO on the X-Series touchscreen HMI: one interface for every test, every system, every brand.

Built for the operator.

Fully automatic testing.

Guided sequences take the component from mounting to final verdict in minutes, with no guesswork. Live actuation signals for advanced analysis and R&D work.

Fast by design.

Complete test cycles measured in minutes, not hours, e.g. **4 Bosch CRDi injectors tested and coded in 8–10 minutes, Delphi C4i in 10-15 minutes, 4 GDi injectors tested and coded in 8–10 minutes; pumps in 5–15 minutes.**

Clear, sellable reports.

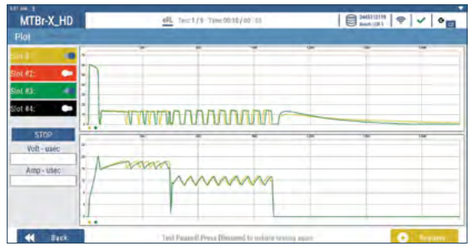
Color-coded pass/fail results per component, a single-page quick report for the customer or a multi-page analytic report for the specialist. Pick your slots with a tap, print or share on the spot, tablet-easy, workshop-fast, and simply incomparable to anything else in the market.

Real-time oscilloscope.

Live actuation signals for advanced analysis and R&D work.



The fail-track color report, results your customer understands at a glance.



The live oscilloscope view, every actuation, every microsecond.



Every connection, mapped on screen: numbered points, exact part numbers and adapters for each component, mount, connect and test with zero guesswork.

One database. Every system.

Complete OE & CZ database.

Test plans for all makes and types, continuously updated:

- **CRDi** - Common Rail Diesel Injectors
- **CRp** - Common Rail Diesel Pumps
- **EUi/EUp** - Electronic Unit Injectors & Pumps
- **HEUi** - Hydraulic Electronic Unit Injectors
- **GDi** - Gasoline (High & Low Pressure) Injectors
- **GDp** - Gasoline Rotary & Piston High Pressure Pumps
- **Marine & Large Engine components**

Your tests, your rules.

User-defined test plans and developer-level customization alongside the factory database, shared to all your Carbon Zapp machines effortlessly.

Machine self-diagnosis.

AZO continuously monitors all the critical points of the bench, sensors, pressures, actuation circuits, and reports the machine's own health before it can affect your results.

Always connected.

On-board Wi-Fi and LAN connectivity for updates, printing, sharing and the live CloudX link.

Always current.

Software and database updates delivered over the air, powered by CloudX (see next spread).

cloudx.carbonzapp.com
carbonzapp.com

CloudX

Your machines, beyond the workshop walls.

CloudX is Carbon Zapp's IoT cloud platform the natural extension of AZO.

Every connected bench becomes part of a secure cloud environment where data, updates, features and support flow both ways. Fully Industry 4.0 compliant, CloudX means the machine you buy today keeps gaining capabilities tomorrow, without a single technician visit.

What CloudX does for your business

Live updates.

Database and software updates arrive online, automatically, your bench always tests to the latest specifications.

Remote monitoring & reports.

Check machine status and generate, file and print test reports from anywhere, on any device.

Complete cloud backup.

Machine data, settings and test history safely mirrored in the cloud, recover everything in minutes.

Fleet management.

Run one bench or twenty: manage every machine, user and location from a single account.

Features on demand.

Activate new capabilities instantly, online, forever or on a Pay-As-You-Go basis, and invest only when the work demands it.

Smart maintenance.

Notifications on your phone for oils, filters and service intervals, before downtime finds you.

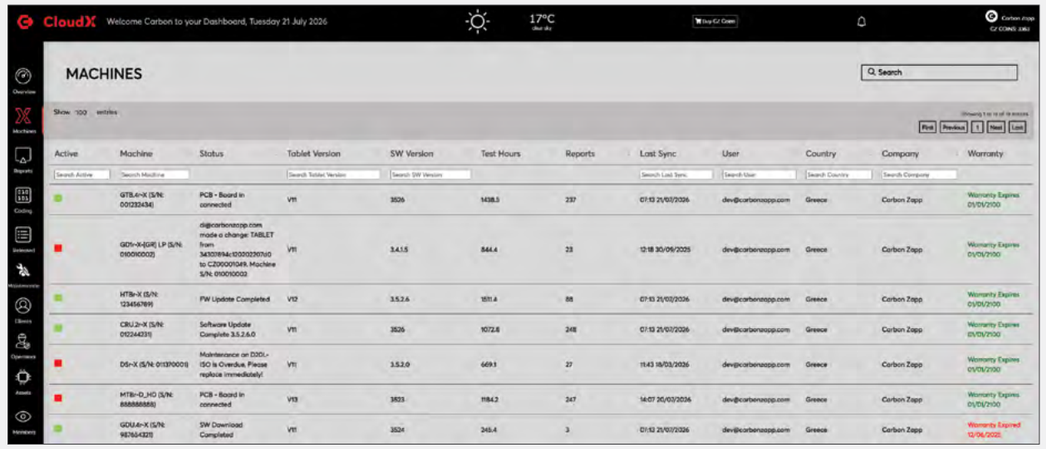
24/7	1 Account	Minutes
Machine visibility from any device	Your entire machine fleet	To activate a new feature online



CloudX
is Carbon Zapp IoT Cloud computing Platform.



The CloudX overview at cloudx.carbonzapp.com Your job in numbers.

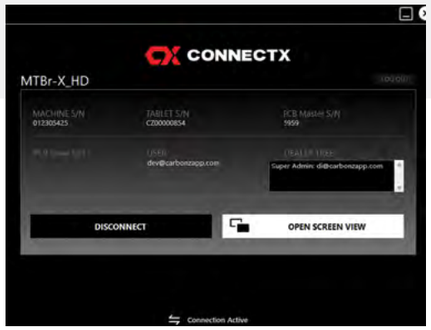


The CloudX fleet dashboard at cloudx.carbonzapp.com Every machine, status and info in one secure view.

EXPERT FEATURES

CloudX unlocks three expert-level extensions ConnectX, API Integration and the Repair Kit Solution (RKS).





A ConnectX session in progress: the machine's screen, live, wherever support happens to be.

ConnectX

Support at the speed of a click.

ConnectX gives authorized eyes and hands on the machine, remotely and per account. When something needs attention, whether a question, a setting or a fault, Carbon Zapp support, your distributor or your own fleet manager can securely connect to the bench and act immediately. No travel, no waiting, no lost workdays.

Instant troubleshooting.

Direct remote assistance from the Carbon Zapp factory or your local distributor, right on the machine screen.

Per-account access.

Connections are permission-based and tied to accounts: you decide who can reach which machine.

Fleet control.

Fleet managers reach any machine in their network to monitor, adjust, update or resolve, from anywhere.

Minimum downtime.

Most issues are resolved in a single remote session. The bench is back to testing the same day.

API Integration

Your test data, in your business systems.

Test data shouldn't stop at the bench. The CloudX API bridges your machines to your own IT environment, so results, reports and machine information flow automatically into the systems your business already runs on: ERP, workshop management, quality and warranty platforms.

Automated workflows.

Test results attach themselves to job cards, invoices and warranty files. No re-typing, no lost paperwork.

Built for scale.

Ideal for remanufacturers, OE service networks and multi-site operations with existing infrastructure.

Secure by design.

Data travels over the same secure cloud environment that powers CloudX.



From the test chamber to your ERP: one connected data path, no manual steps.

Repair Kit Solution

The authorized channel



The RKS area in CloudX: providers publish, workshops access, and every kit is tested the authorized way.

For kit providers

Deploy to your network.

Publish your repair kits, data and test plans directly to your authorized workshops through CloudX.

Control who buys.

Entitlements are managed per user: only your authorized network can access and purchase your kits.

Protect your brand.

Every kit is diagnosed and validated on a Carbon Zapp bench, against your specifications. Quality and warranty stay in your hands.

A complete, branded solution.

From kit deployment to the final test report, the whole workflow carries your brand, a professional, consistent outcome across your entire network.

For workshops

Genuine kits, on the bench.

Access authorized repair kits with the matching test plans, right where the work happens.

No more parts hunting.

Become an authorized workshop with any of the available providers for your systems, the right kit is always one click away.

Certified results.

Diagnosis and post-repair testing follow the provider's approved procedure, every repair leaves with proof.

One machine. One platform. The complete ecosystem.

AZO runs the test. CloudX connects the machine. ConnectX keeps it running. The API brings the data home. RKS closes the loop between component, kit and certified repair. Together, they turn a Carbon Zapp bench into a business platform.

Built to Move the Industry Forward.

From advanced testing equipment and connected intelligence to technical expertise and global support, Carbon Zapp brings every solution together in one complete ecosystem.

Let's Connect and Discover Future Opportunities.

Scan the QR Code and easily download the Product Catalog to your phone.

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Scan and Explore
Complete CZ Solutions